

ERD



Figure 3-7 -- A **composite** attribute

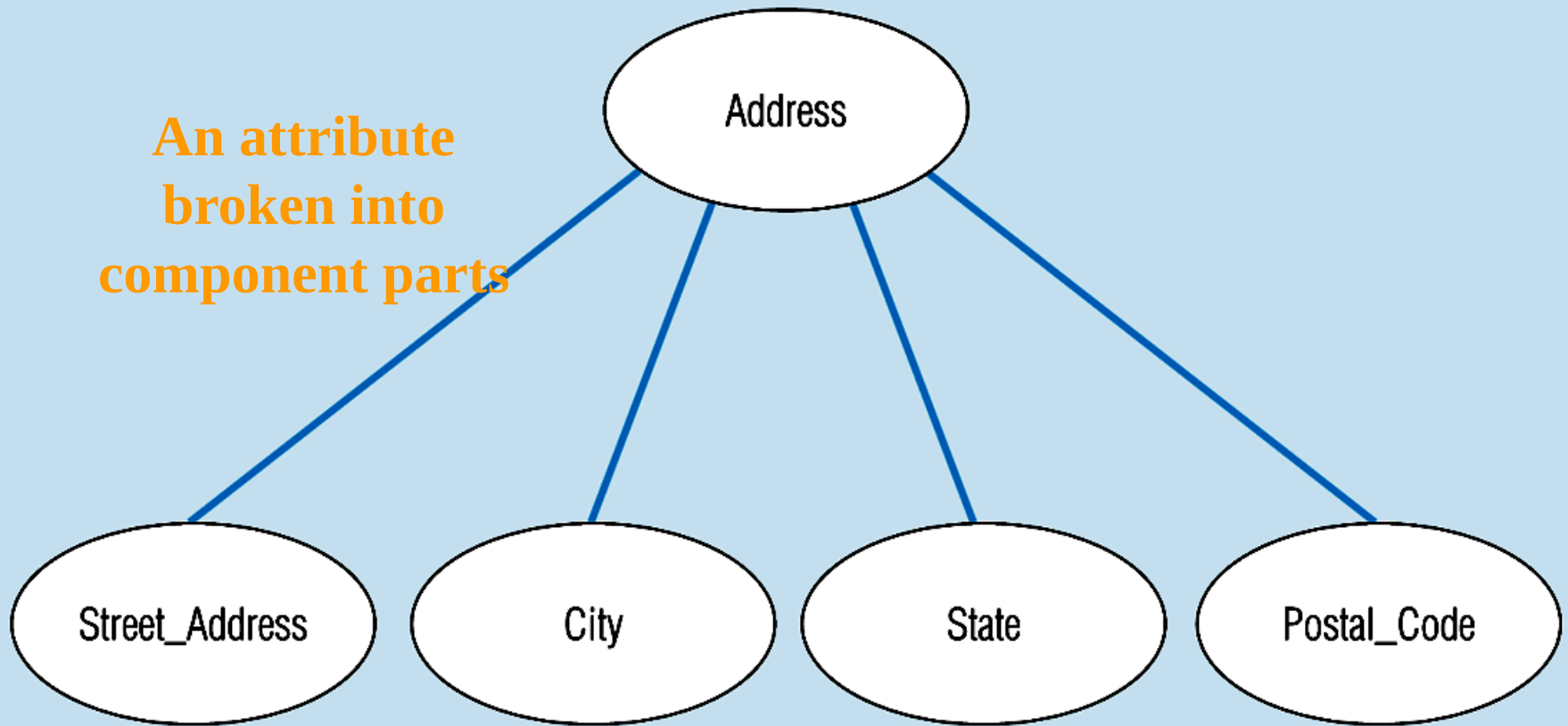


Figure 3-9a – Simple key attribute

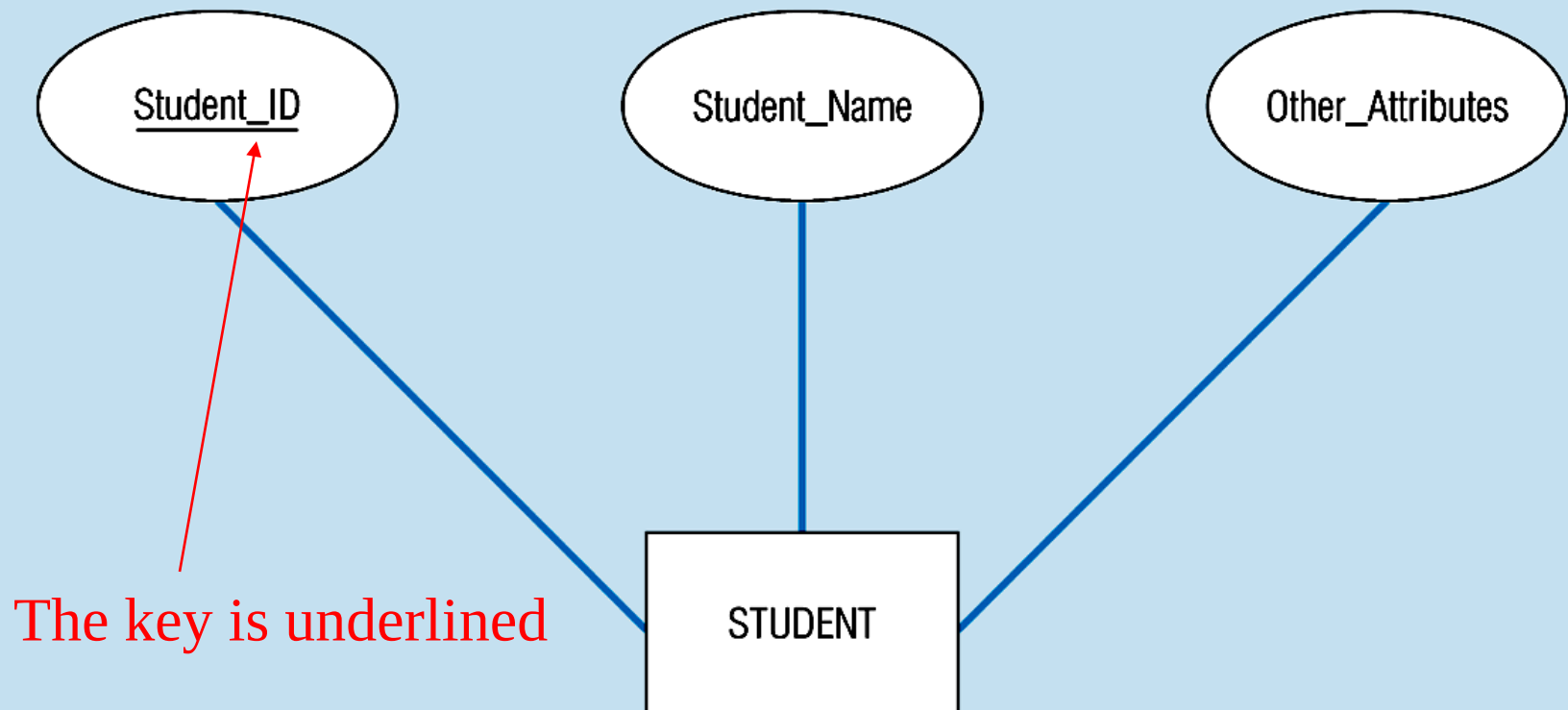


Figure 3-9b -- Composite key attribute

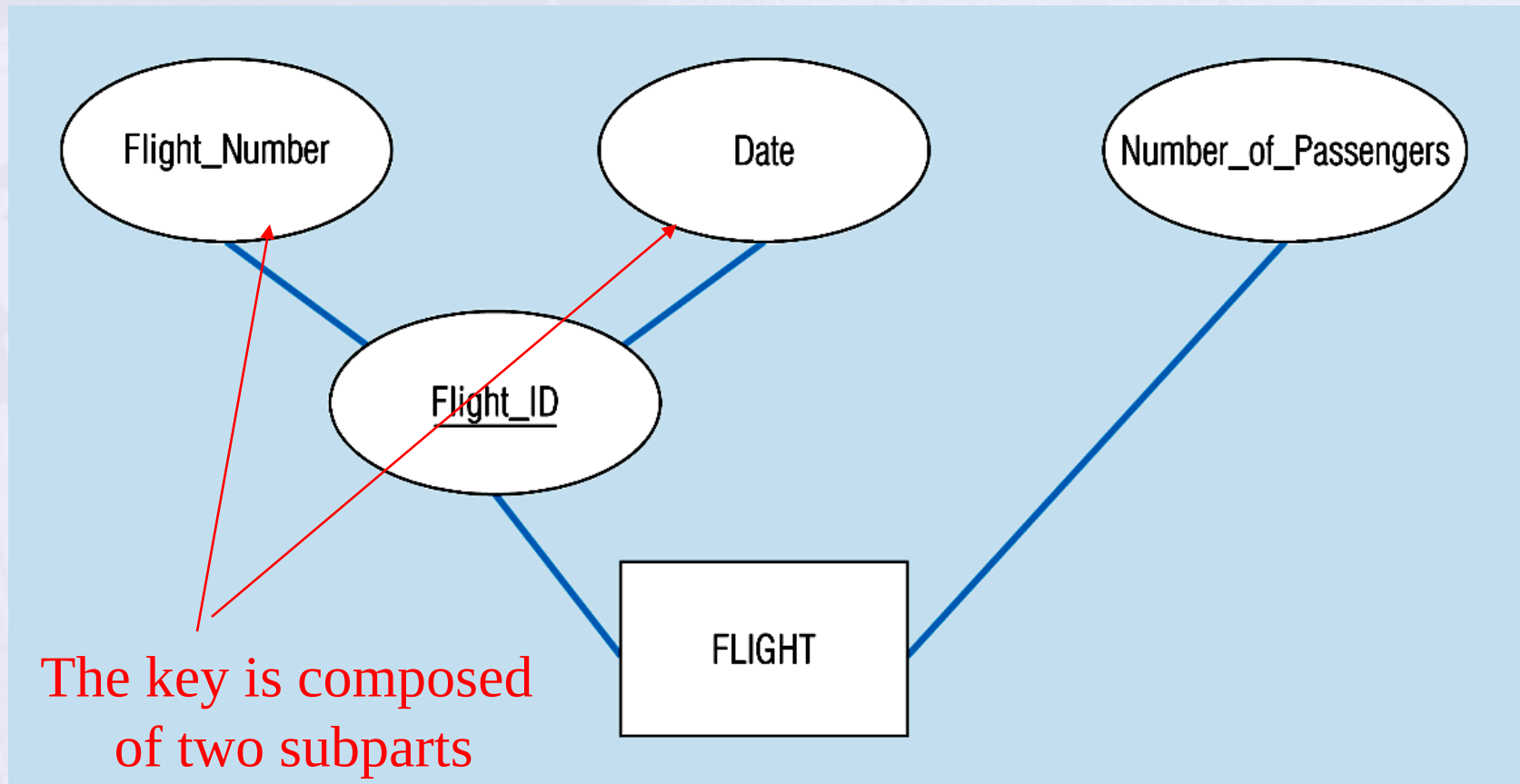


Figure 3-8 -- Entity with a multivalued attribute (Skill) and derived attribute (Years_Employed)

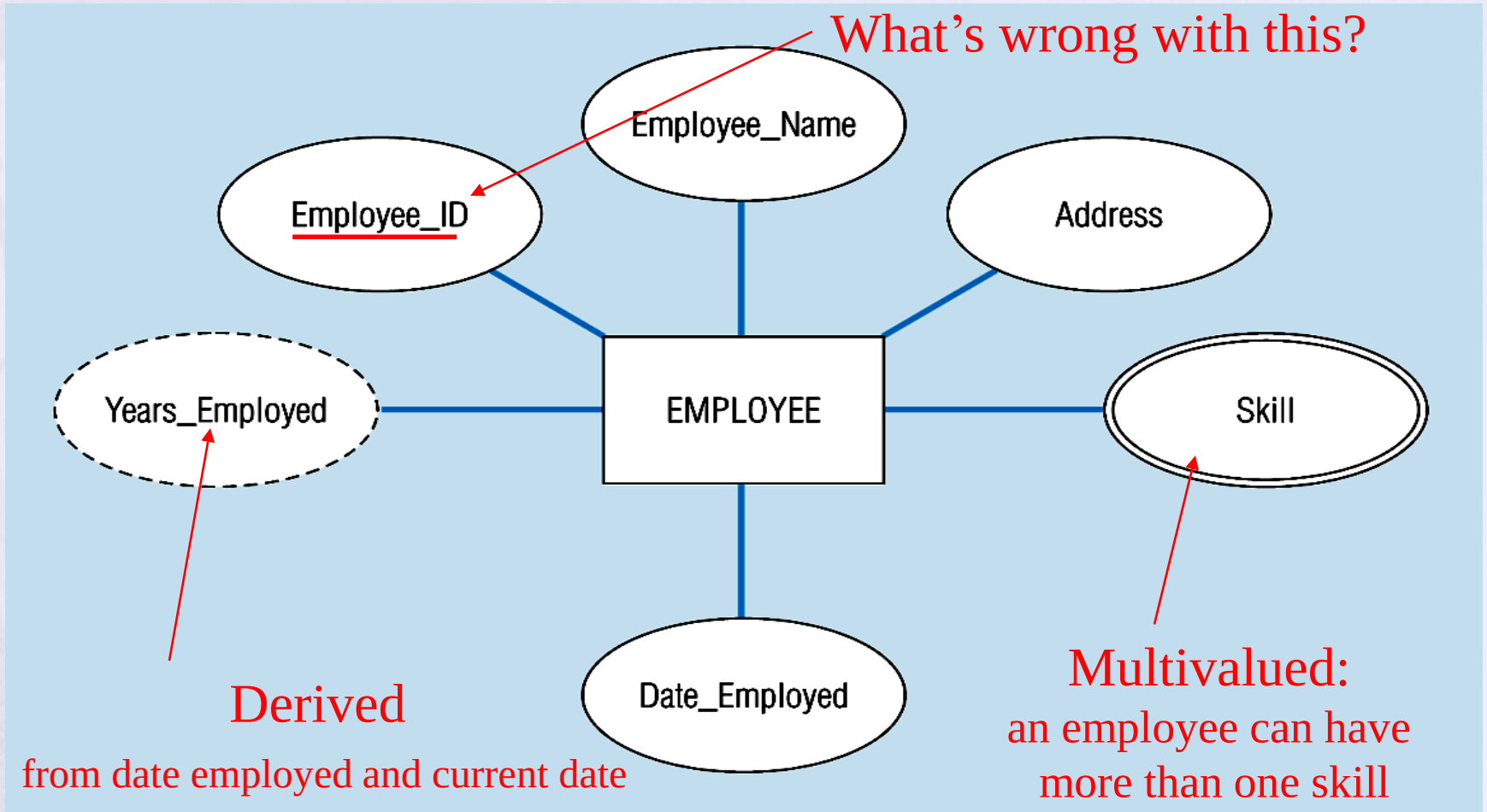
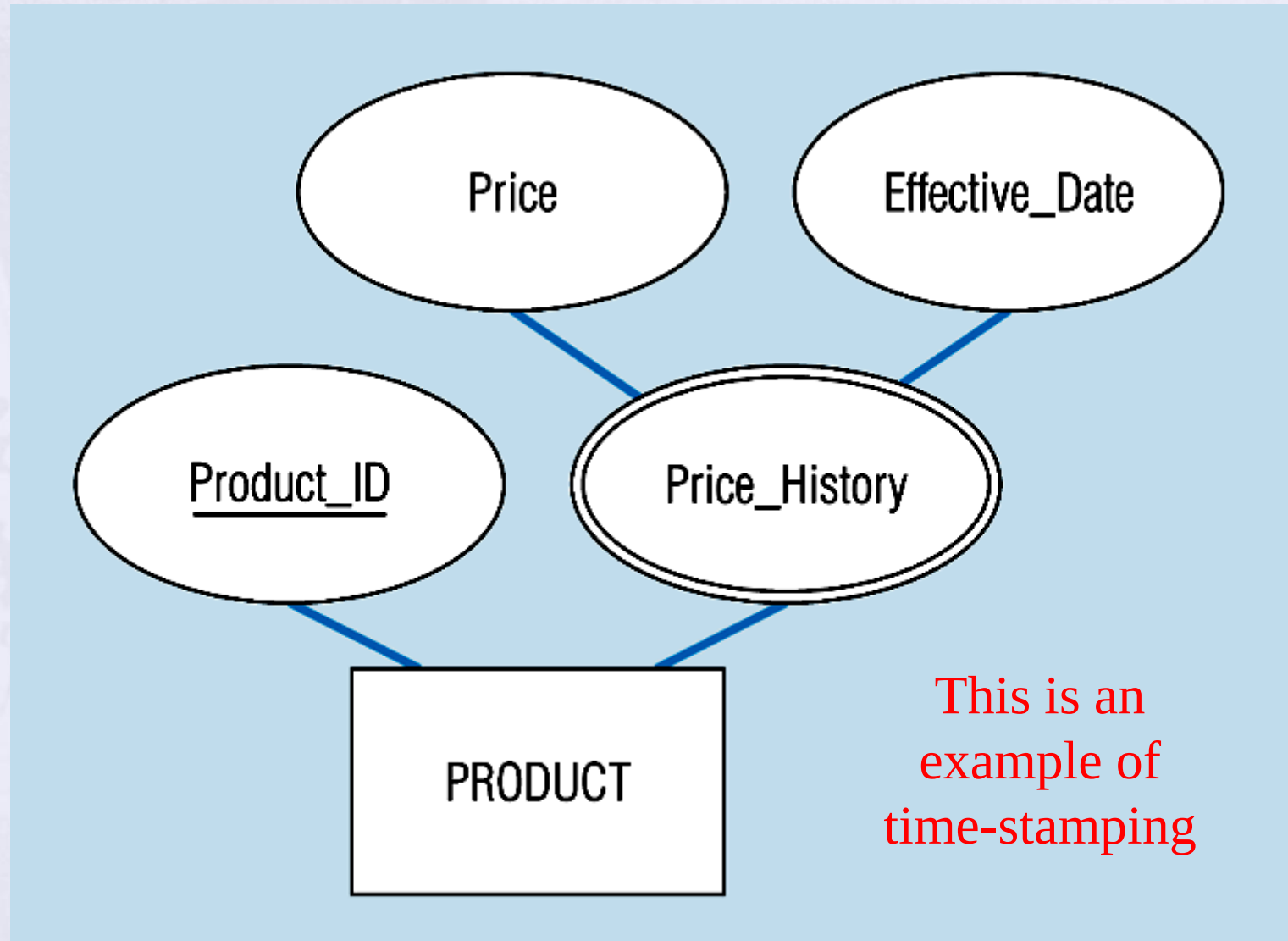


Figure 3-19 – an attribute that is both multivalued and composite



More on Relationships

➤ Relationship Types vs. Relationship Instances

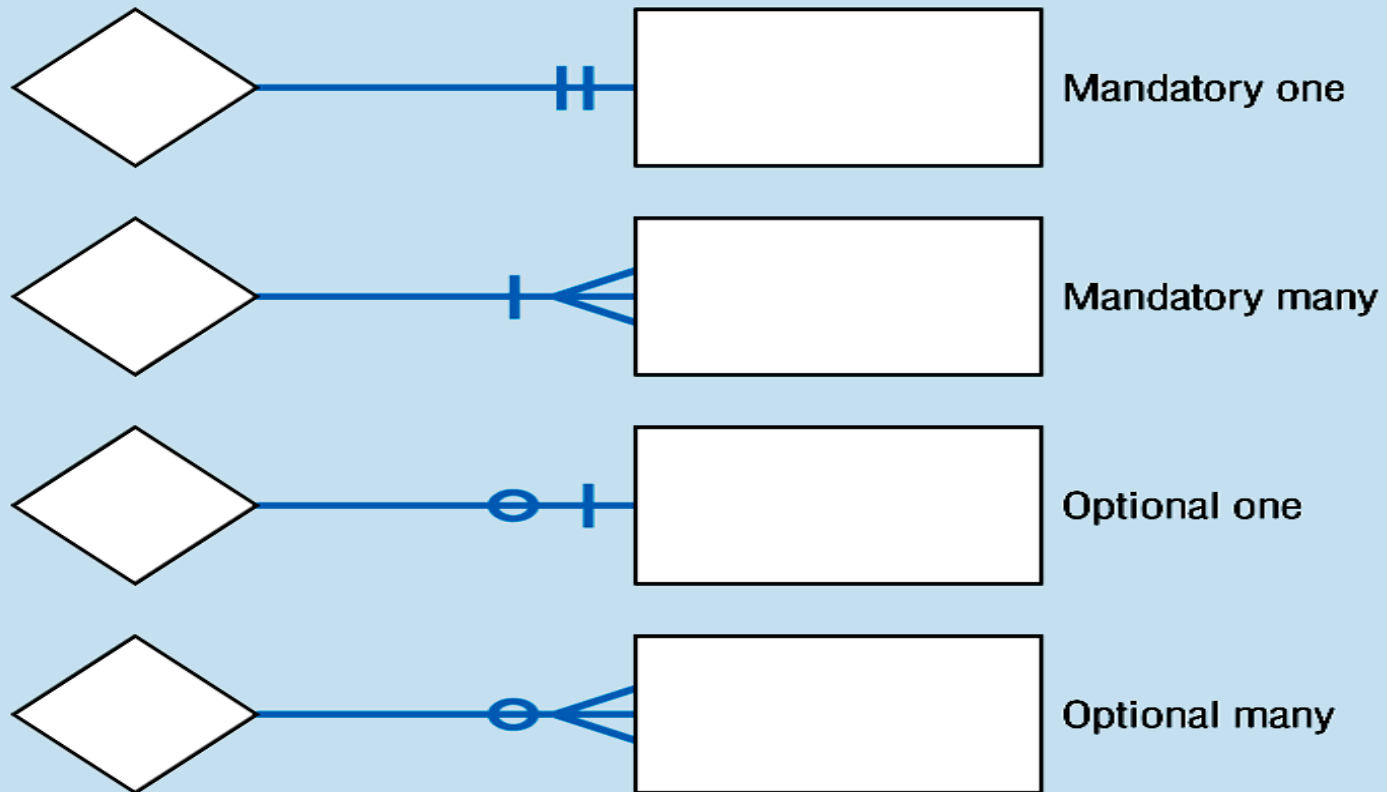
- The relationship type is modeled as the diamond and lines between entity types...the instance is between specific entity instances
- Relationships can have attributes
 - These describe features pertaining to the association between the entities in the relationship
- Two entities can have more than one type of relationship between them (multiple relationships)
- Associative Entity = combination of relationship and entity
 - More on this later

Cardinality of Relationships

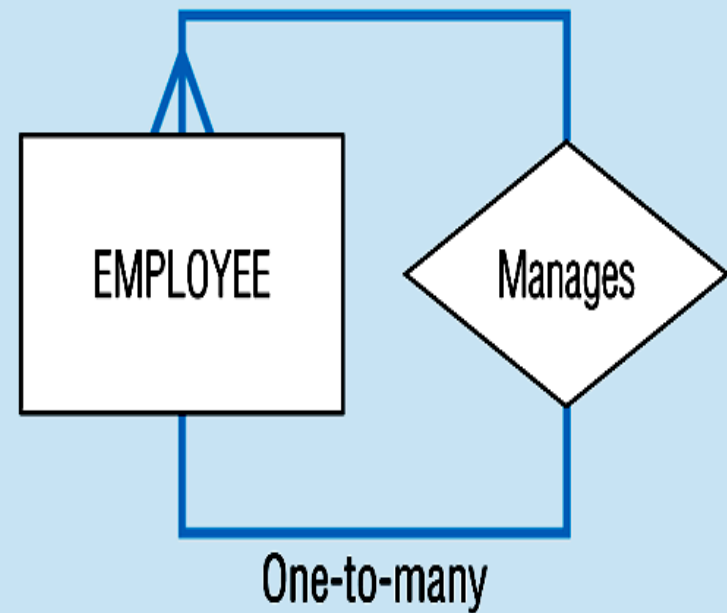
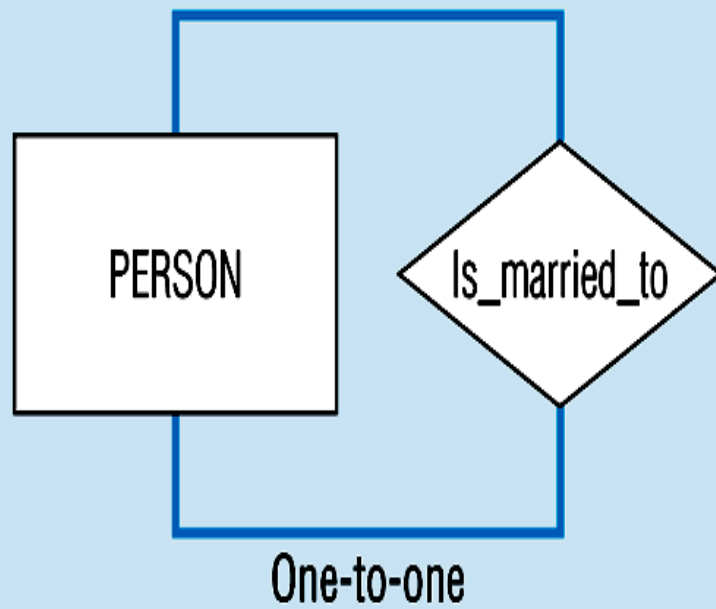
- One – to – One
 - Each entity in the relationship will have exactly one related entity
- One – to – Many
 - An entity on one side of the relationship can have many related entities, but an entity on the other side will have a maximum of one related entity
- Many – to – Many
 - Entities on both sides of the relationship can have many related entities on the other side

Cardinality – figure 3-2

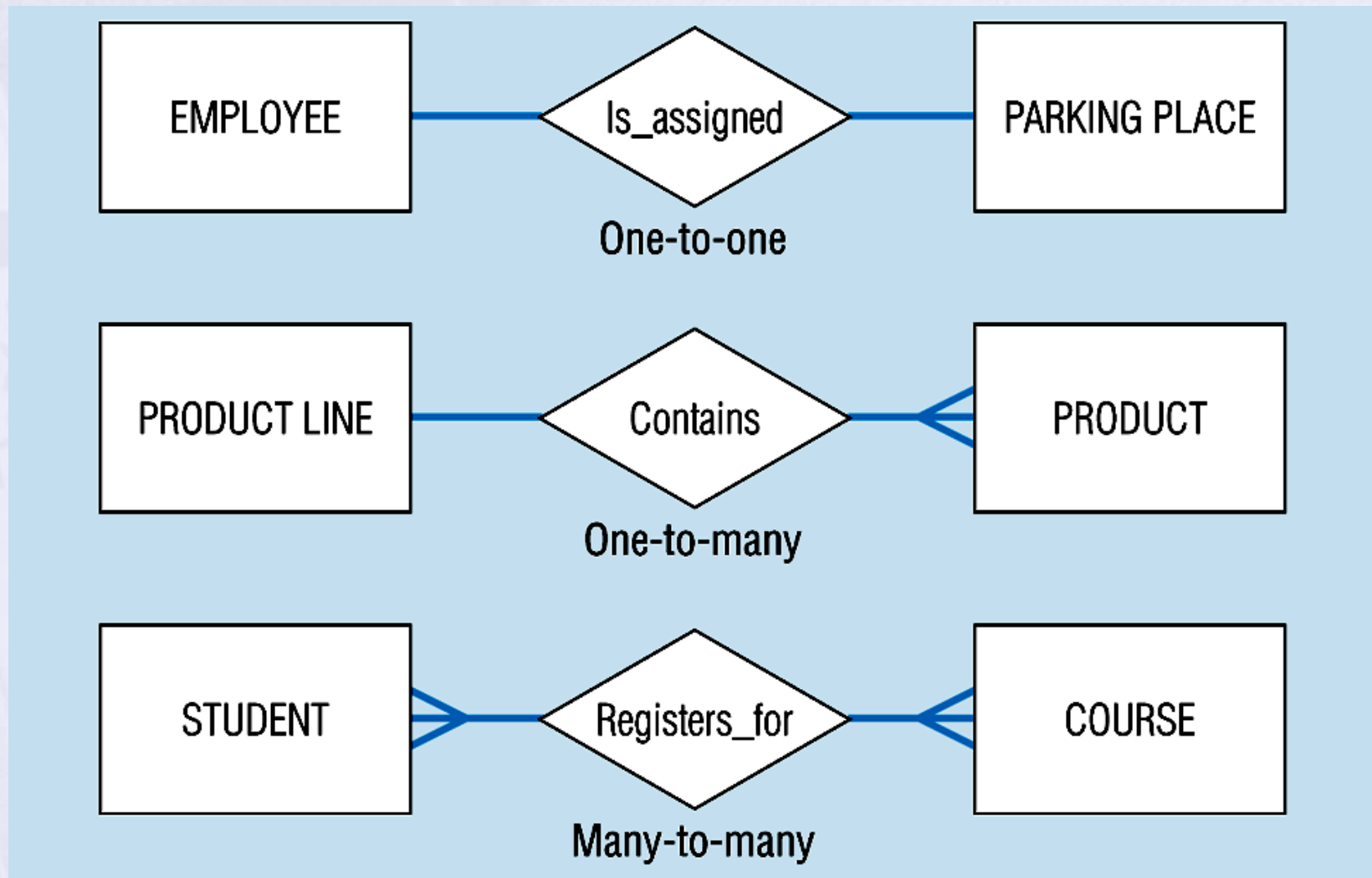
Relationship cardinality



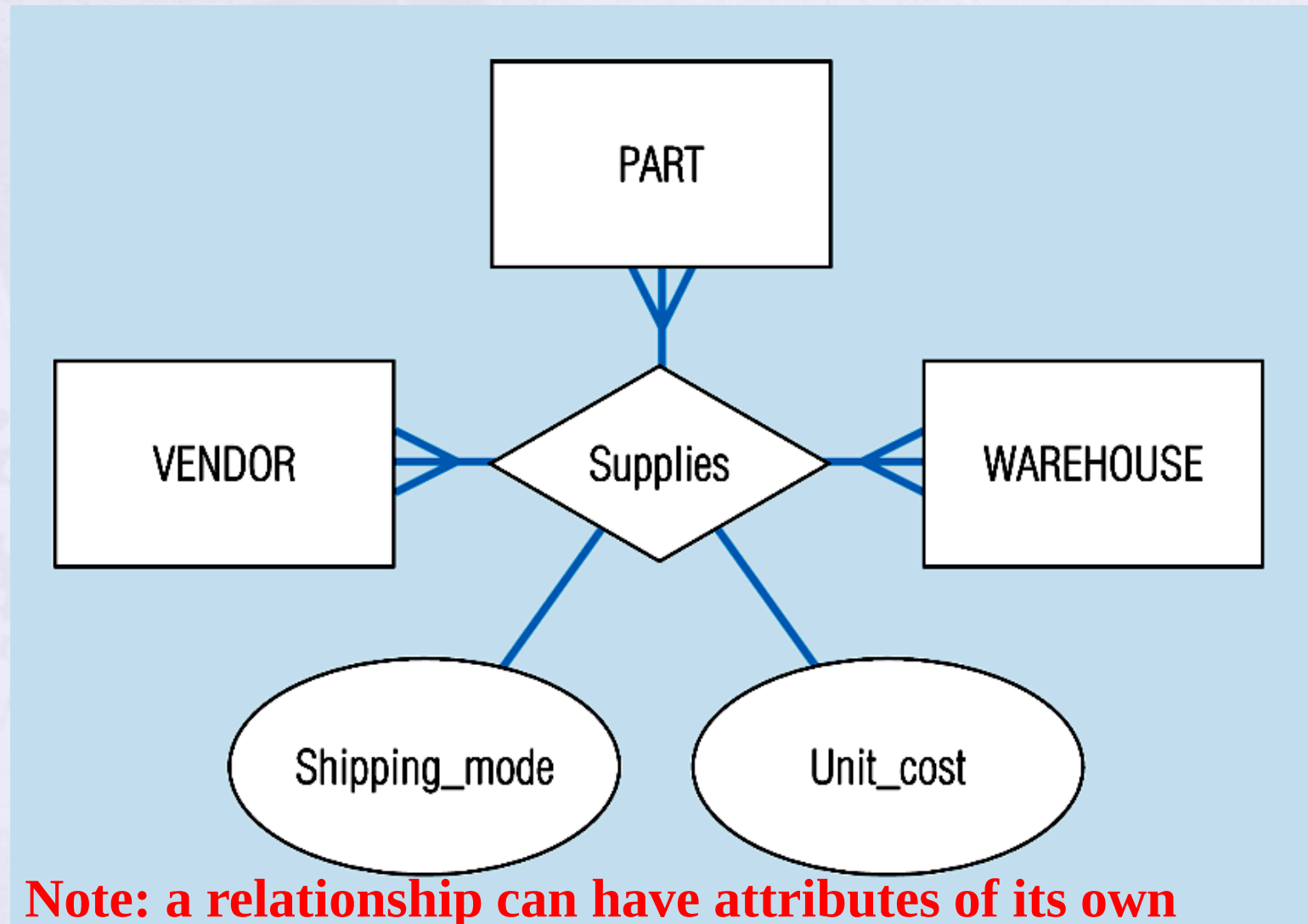
Unary relationships -- figure 3-12a



Binary relationships – figure 3-12b



Ternary relationships –figure 3-12c



Basic relationship with only maximum cardinalities showing –
figure 3-16a



Mandatory minimum cardinalities – figure 3-17a

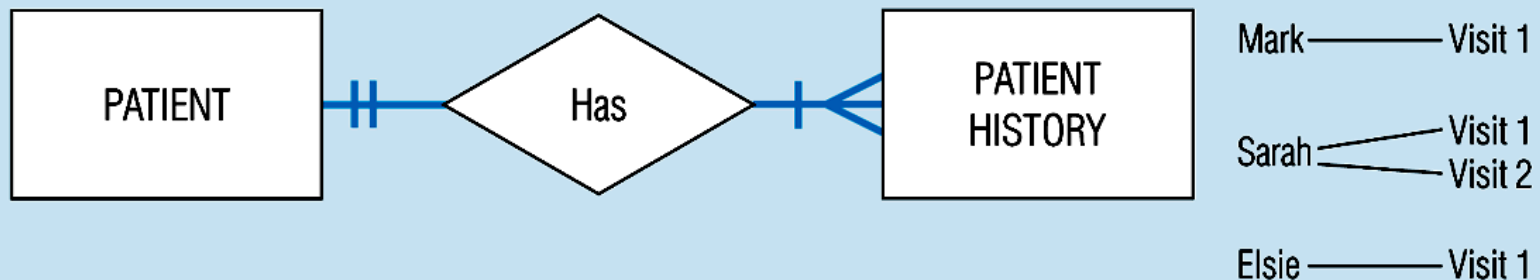


Figure 3-17c
Optional cardinalities with unary degree, one-to-one relationship

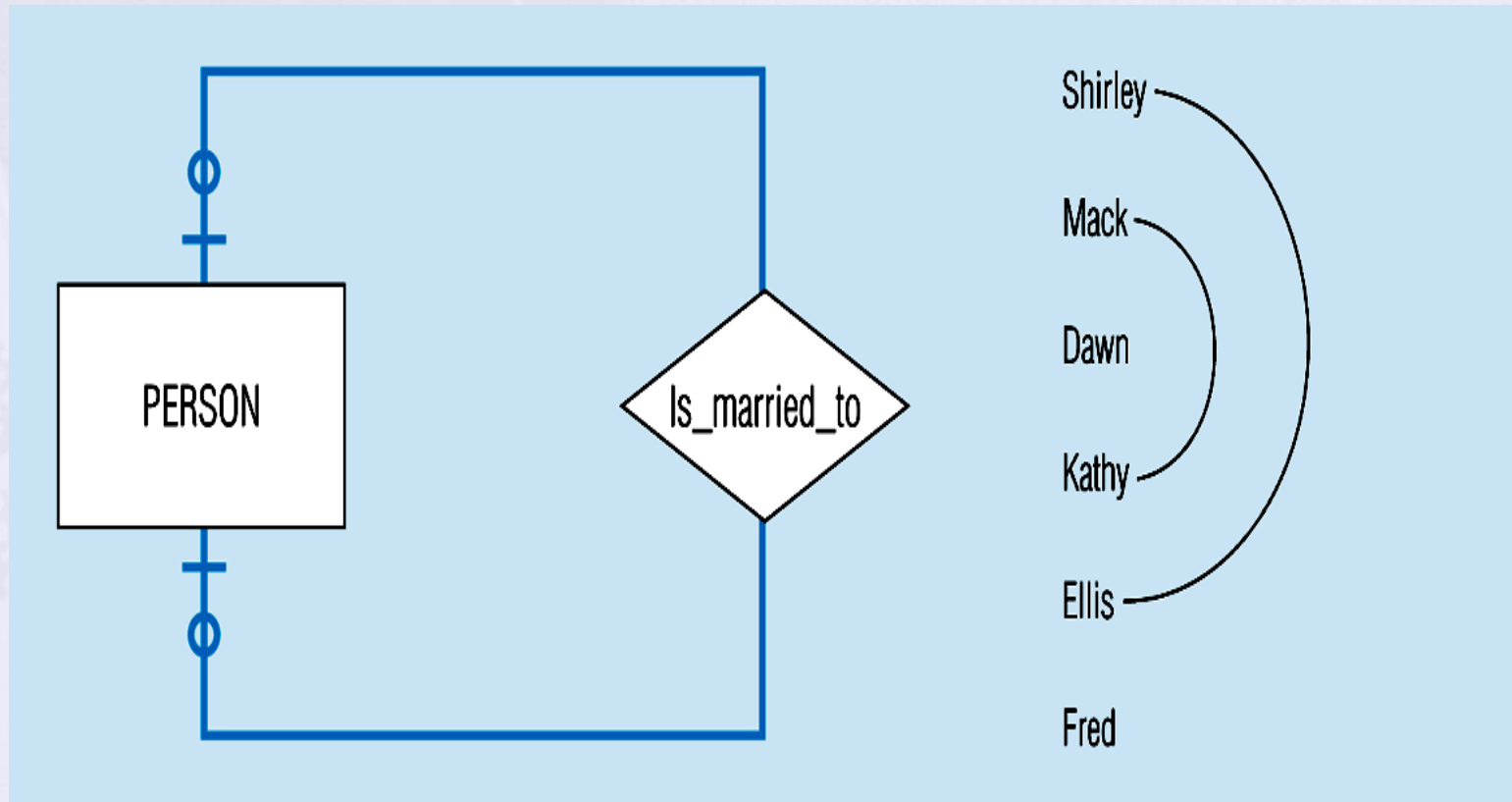
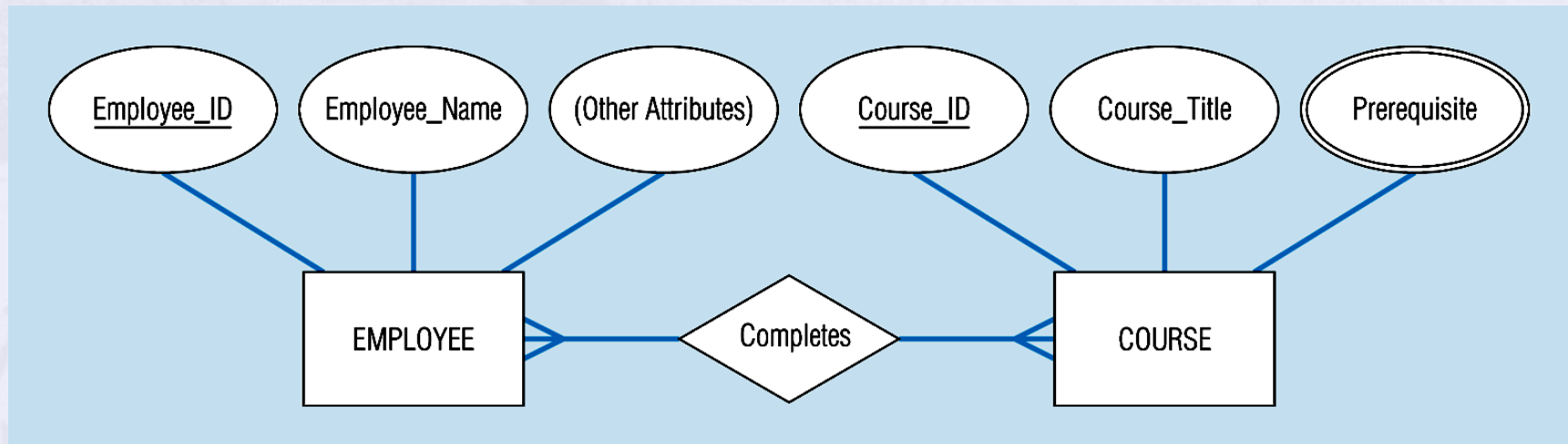


Figure 3-10a Relationship type



3-10b Entity and Relationship instances

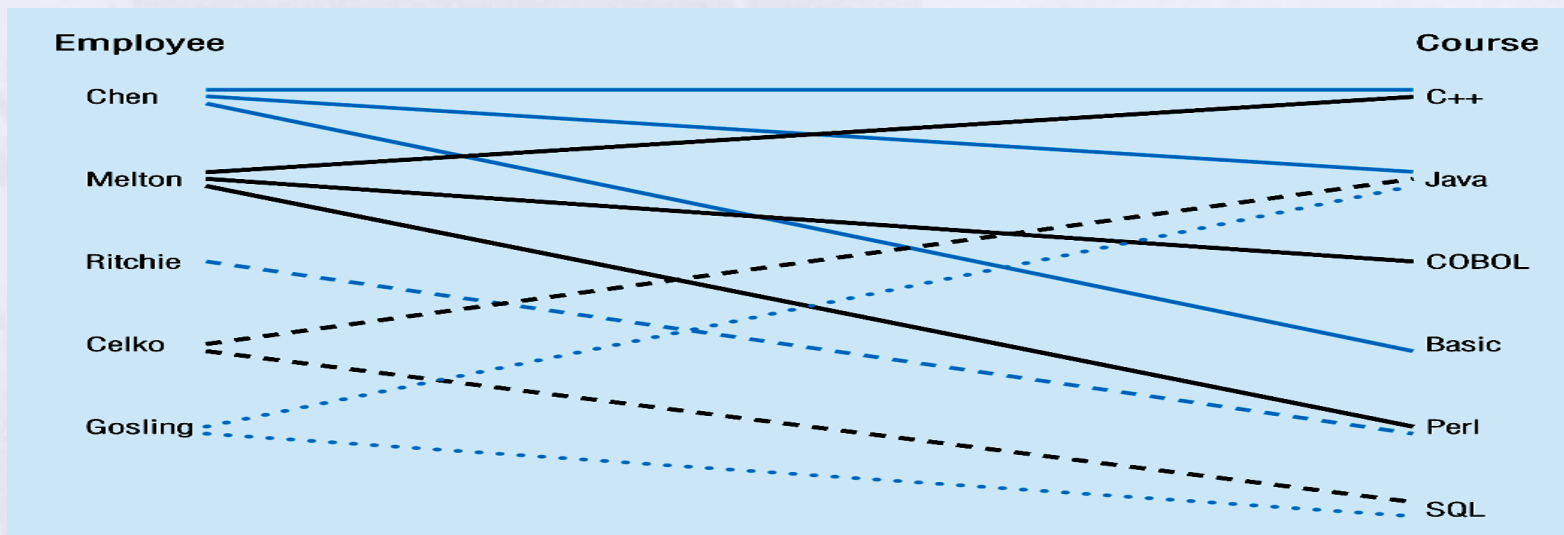


Figure 3-12c -- A ternary relationship with attributes

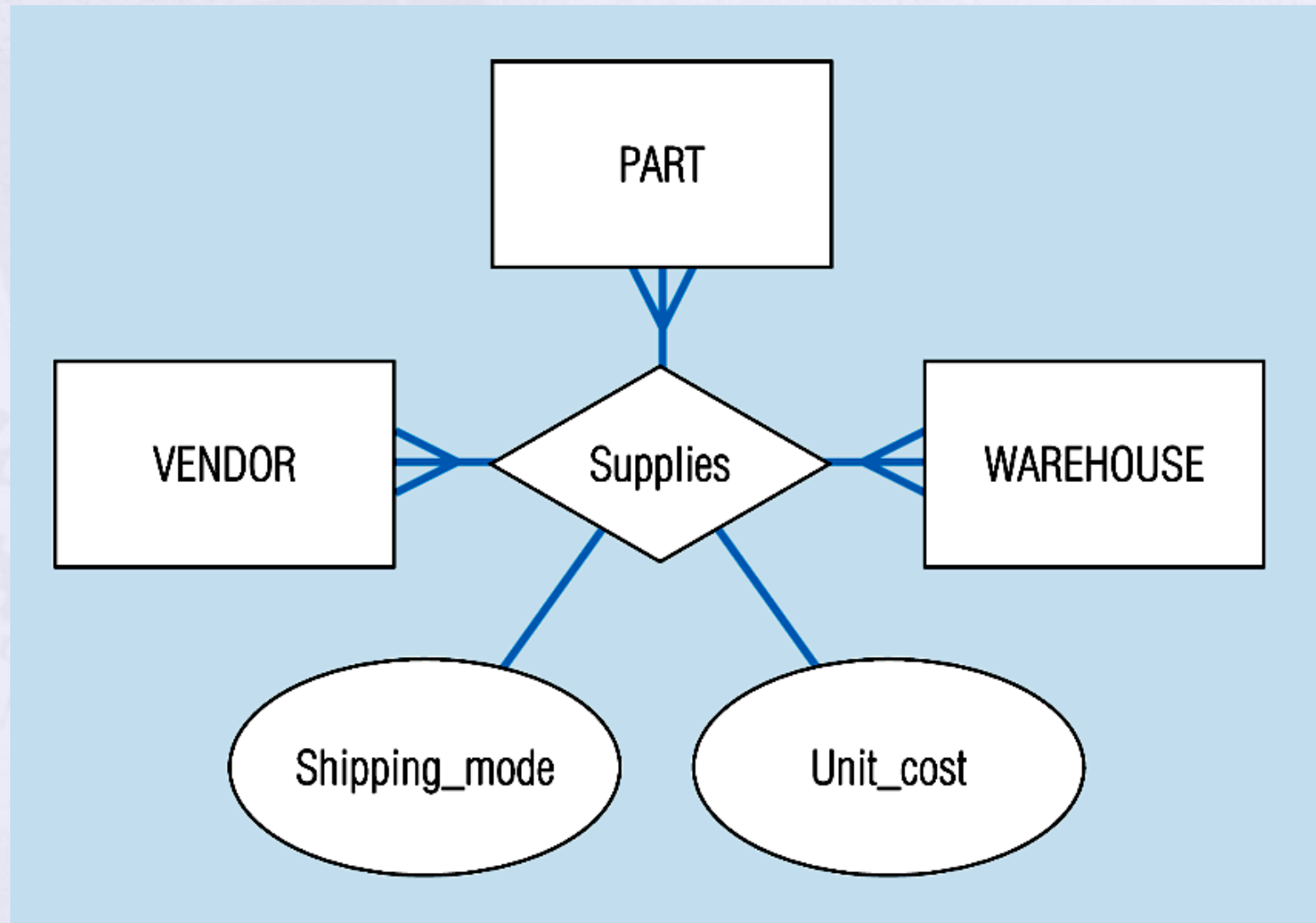
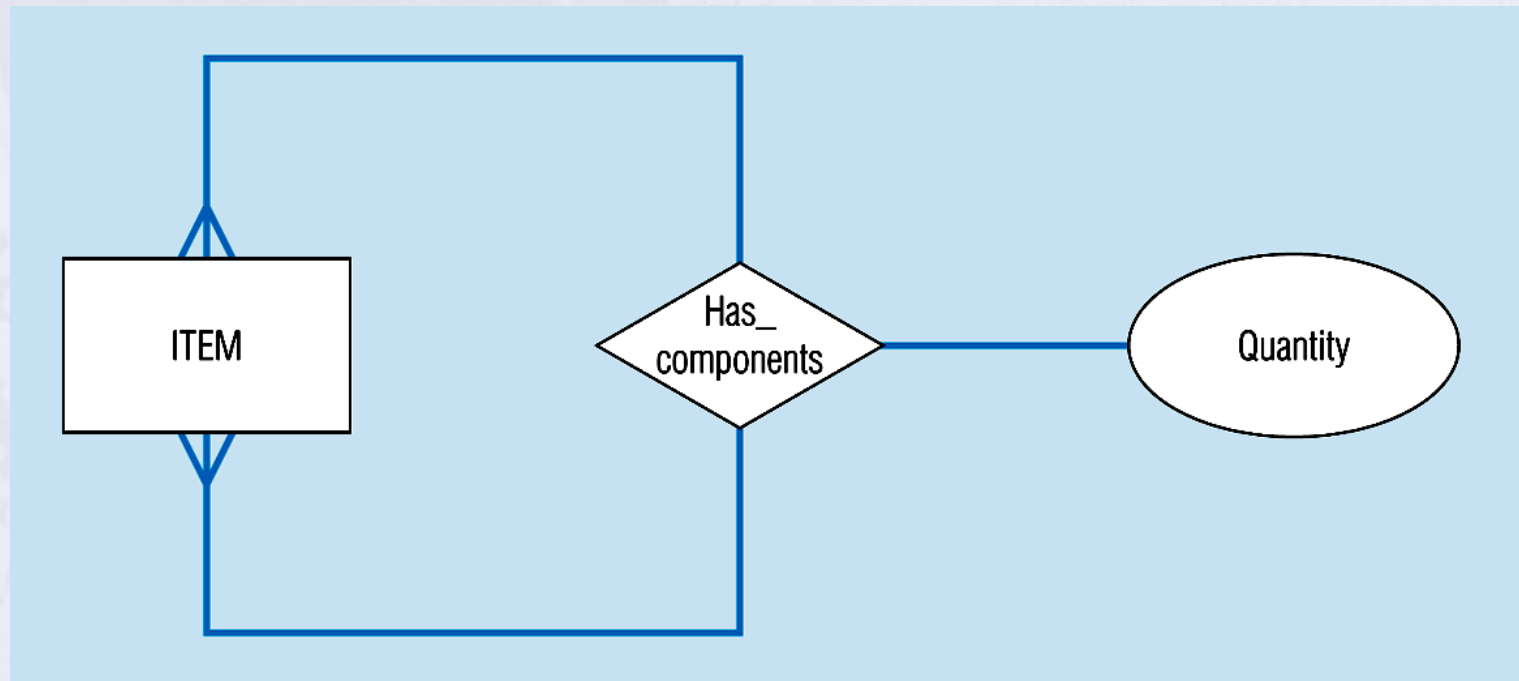


Figure 3-13a A unary relationship with an attribute. This has a many-to-many relationship



Representing a bill-of -materials structure

Examples of multiple relationships – entities can be related to one another in more than one way

Figure 3-21a Employees and departments

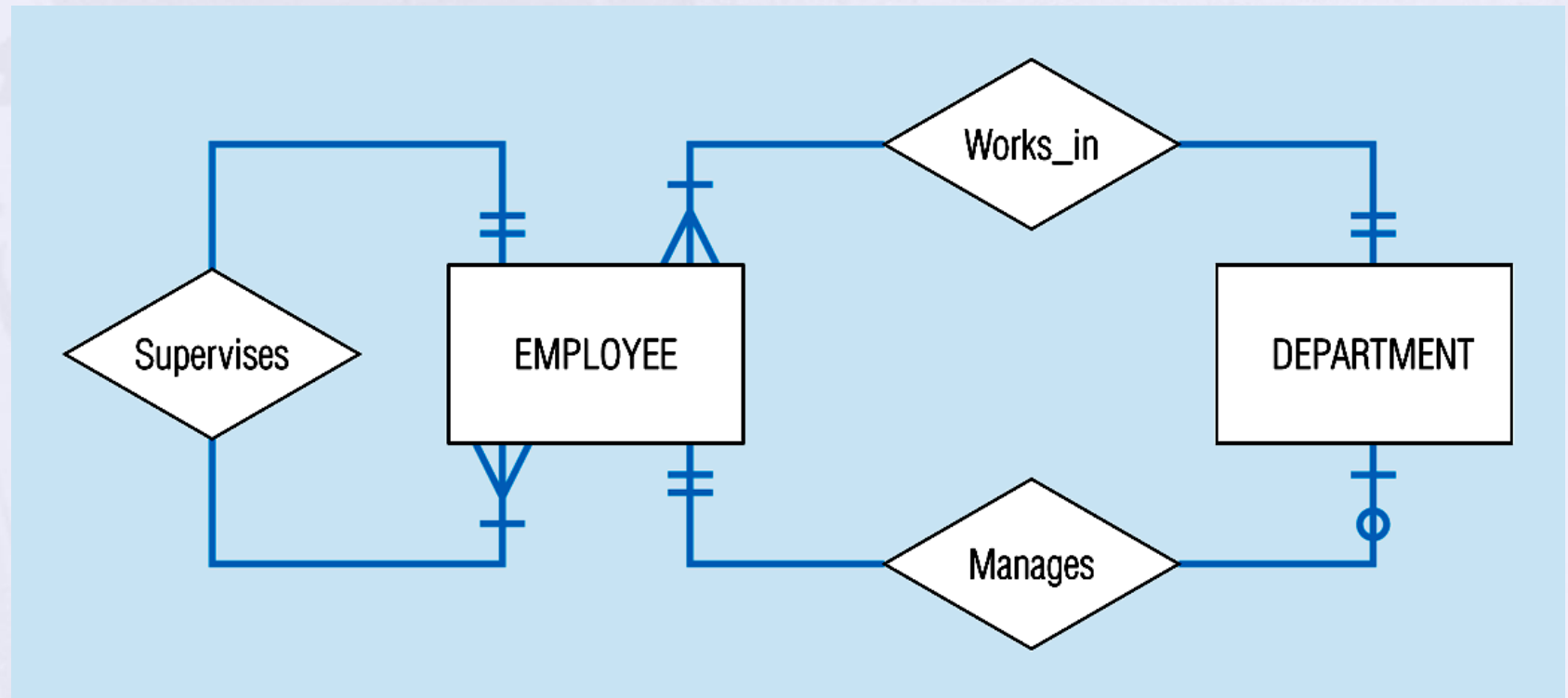
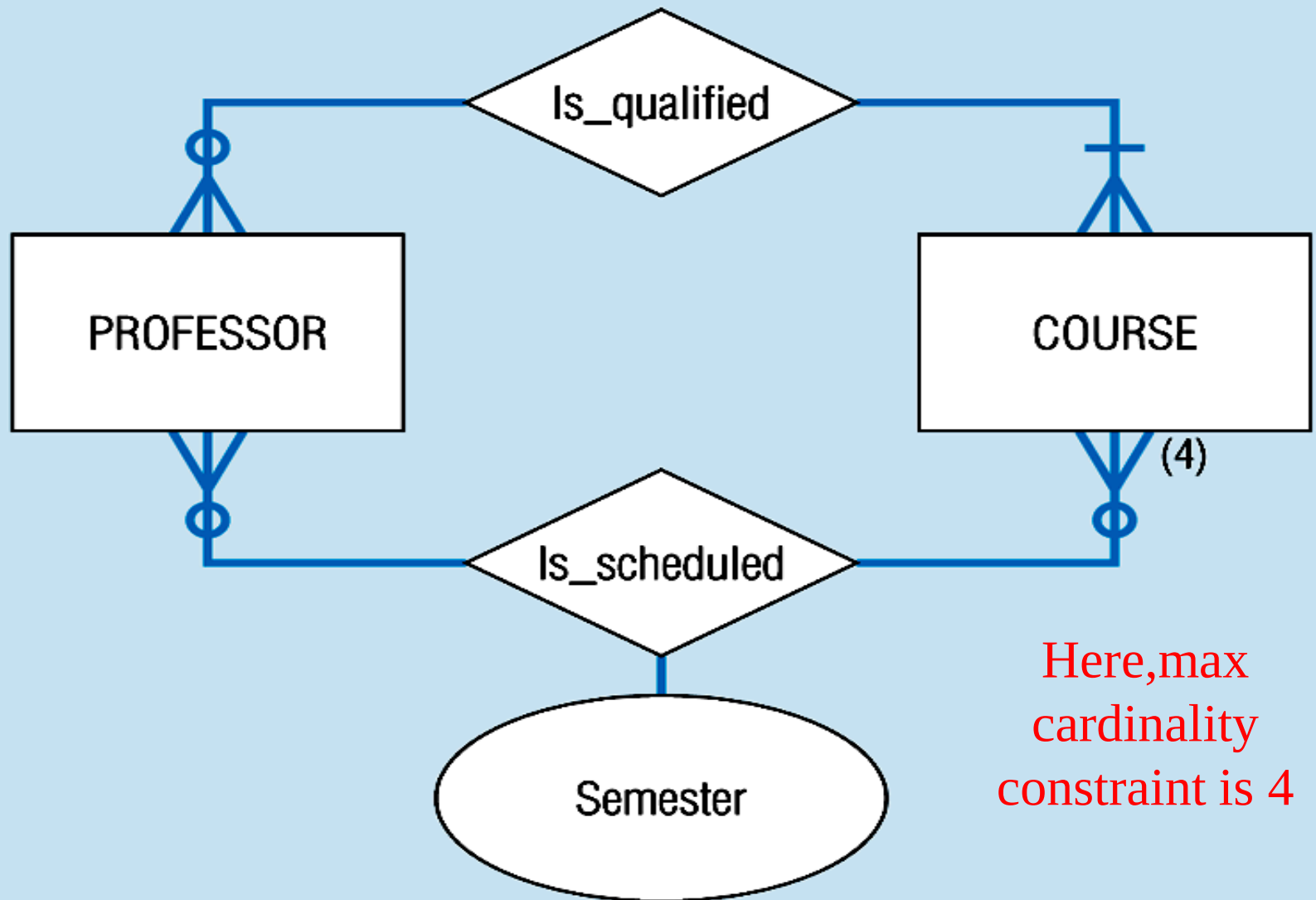
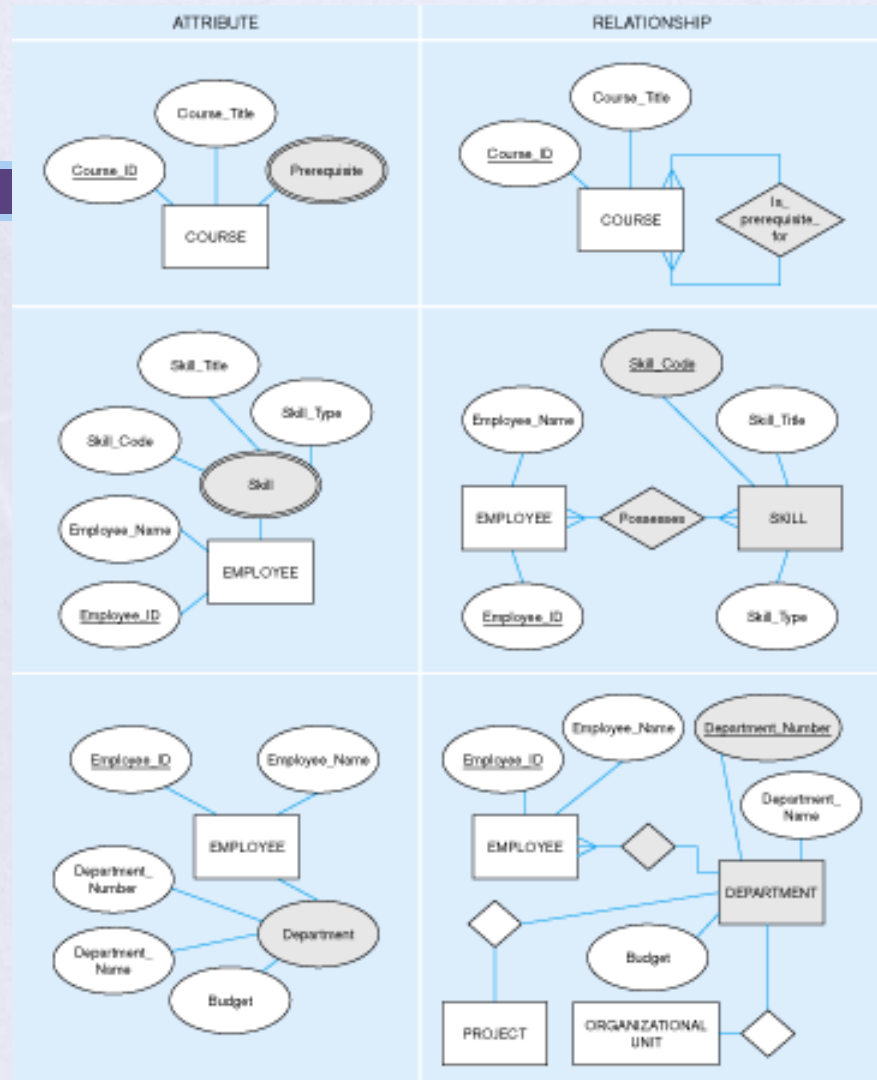


Figure 3-21b -- Professors and courses (fixed upon constraint)



Here, max
cardinality
constraint is 4

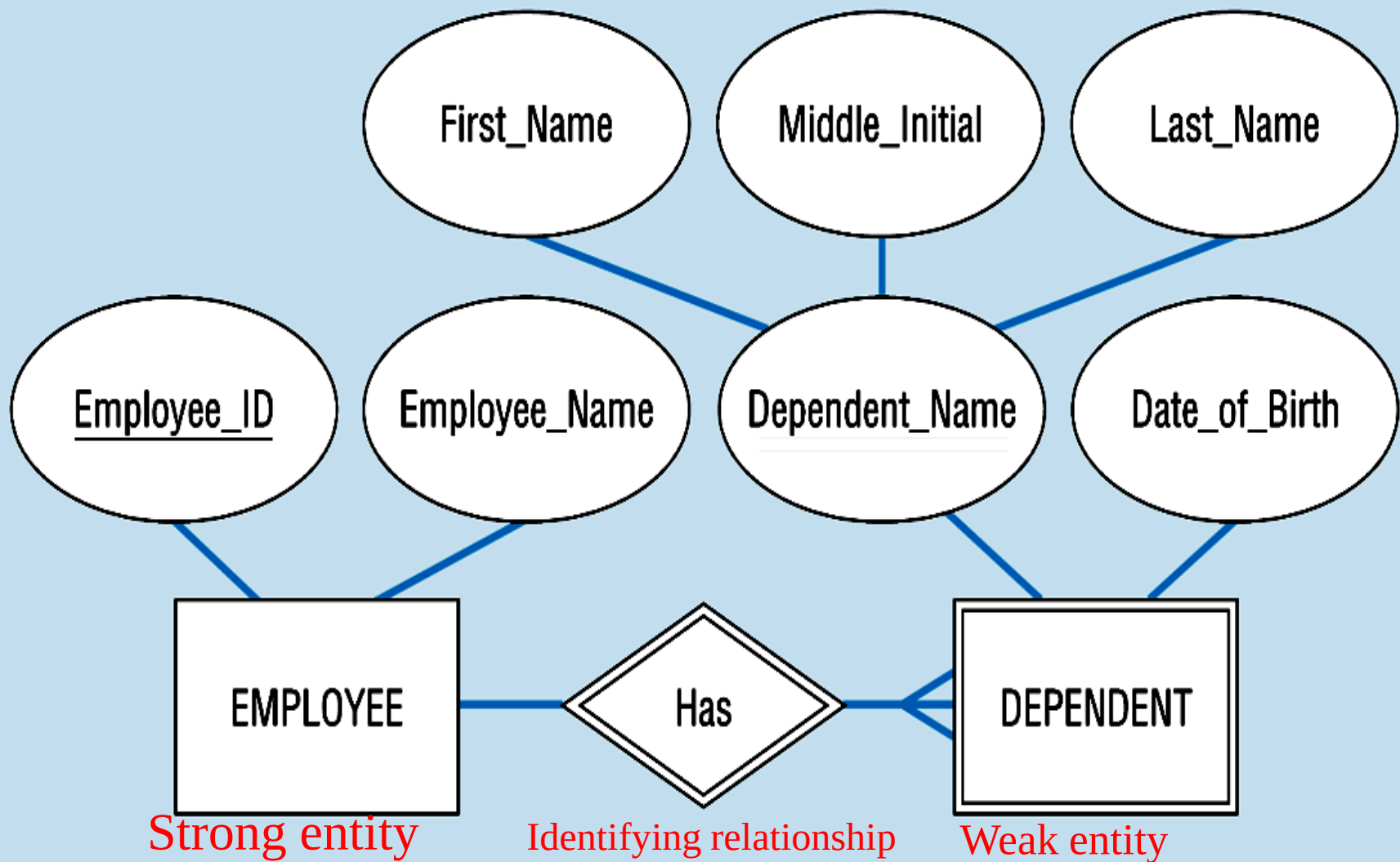
Figure 3-15:
Multivalued attribute
vs. relationship.
Alternative approaches



Strong vs. Weak Entities, and Identifying Relationships

- Strong entities
 - exist independently of other types of entities
 - has its own unique identifier
 - represented with single-line rectangle
- Weak entity
 - dependent on a strong entity...cannot exist on its own
 - Does not have a unique identifier
 - represented with double-line rectangle
- Identifying relationship
 - links strong entities to weak entities
 - represented with double line diamond

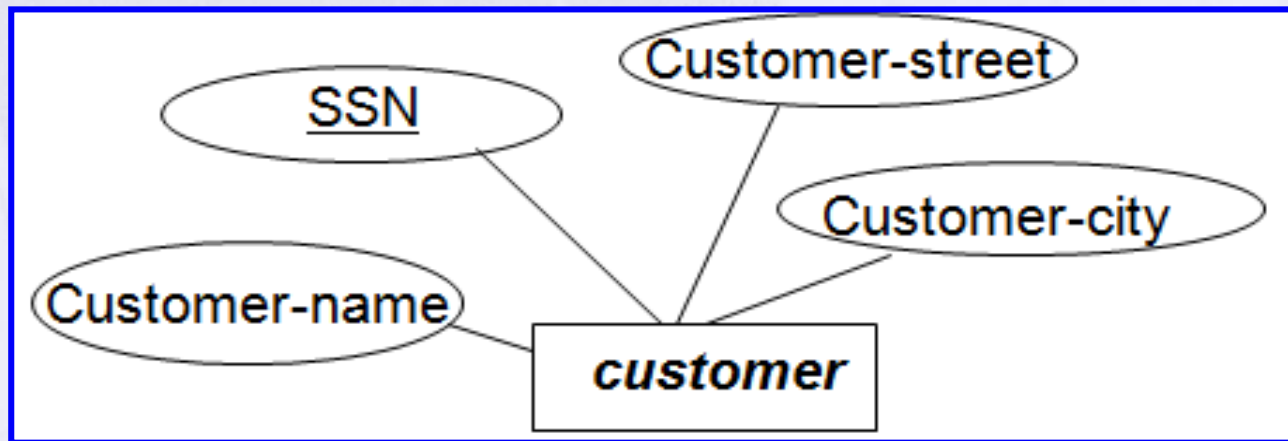
Figure 3-5: Strong and weak entities



Super Key

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- **Superkey**: is a set of attributes whose value can uniquely identify an entity in the entity set.
- A super key contains one or more attributes.



The value of attributes “**SSN**” and “**Name**”, such as **558-36-1234** and **Susan**, can uniquely identify that particular customer in *customer* entity set

Other super keys for the entity set *customer*.

Superkey: (SSN)

Superkey: (SSN, Street)

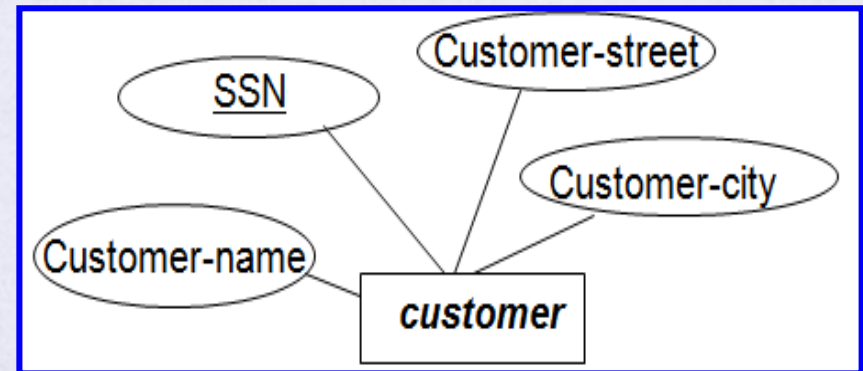
Superkey: (SSN, City)

Superkey: (SSN, Name, Street)

Superkey: (SSN, Name, City)

Superkey: (SSN, Street, City)

Superkey: (SSN, Name, Street, City)



Comparing the Keys:

Super Key	Candidate Key	Primary Key
Broadest Unique Identifier	Is a Subset of Super Key	Is a subset of Candidate Key

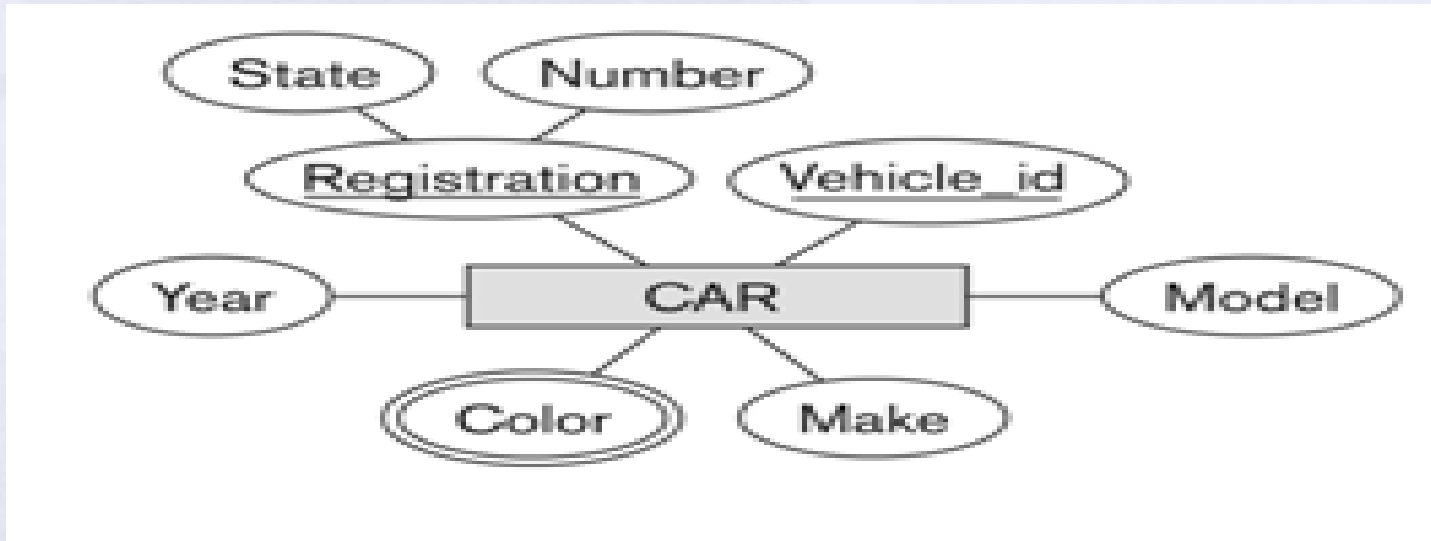


Figure 5-14: Mapping a binary 1:1 relationship

(a) Binary 1:1 relationship

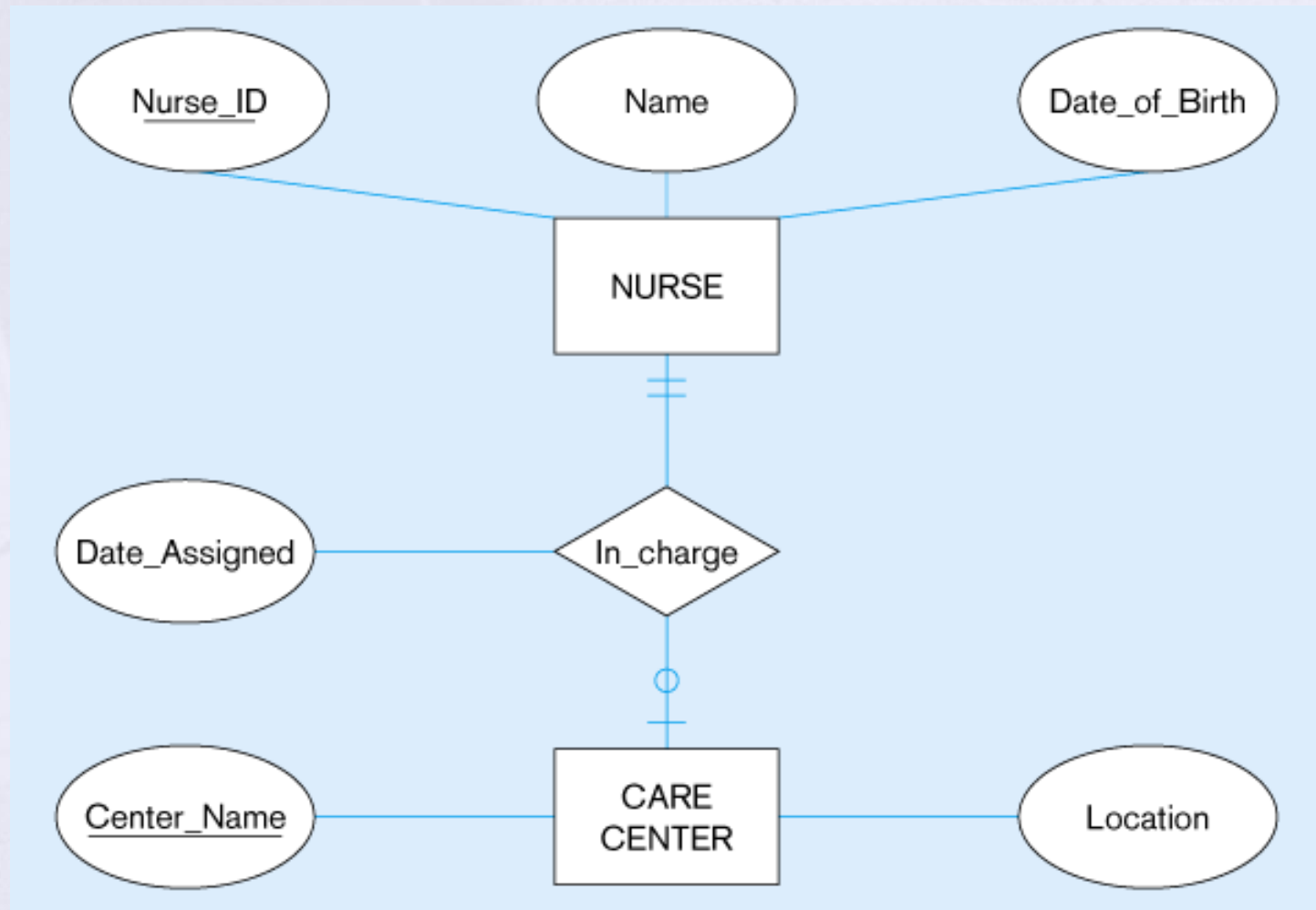


Figure 5-14(b) Resulting relations

NURSE

<u>Nurse_ID</u>	Name	Date_of_Birth
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CARE CENTER

<u>Center_Name</u>	Location	<u>Nurse_in_Charge</u>	Date_Assigned
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Figure 5-12: Example of mapping a 1:M relationship

(a) Relationship between customers and orders

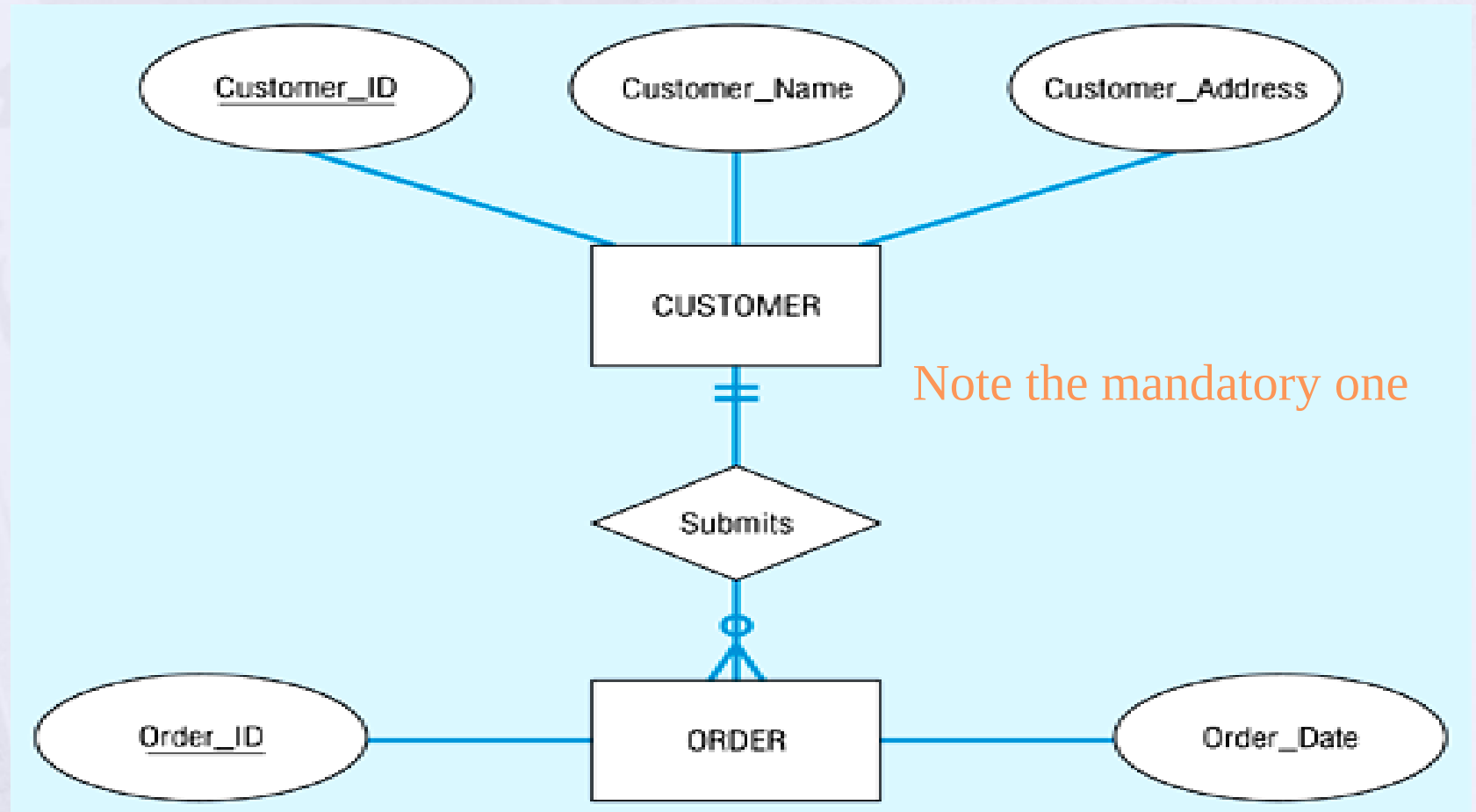
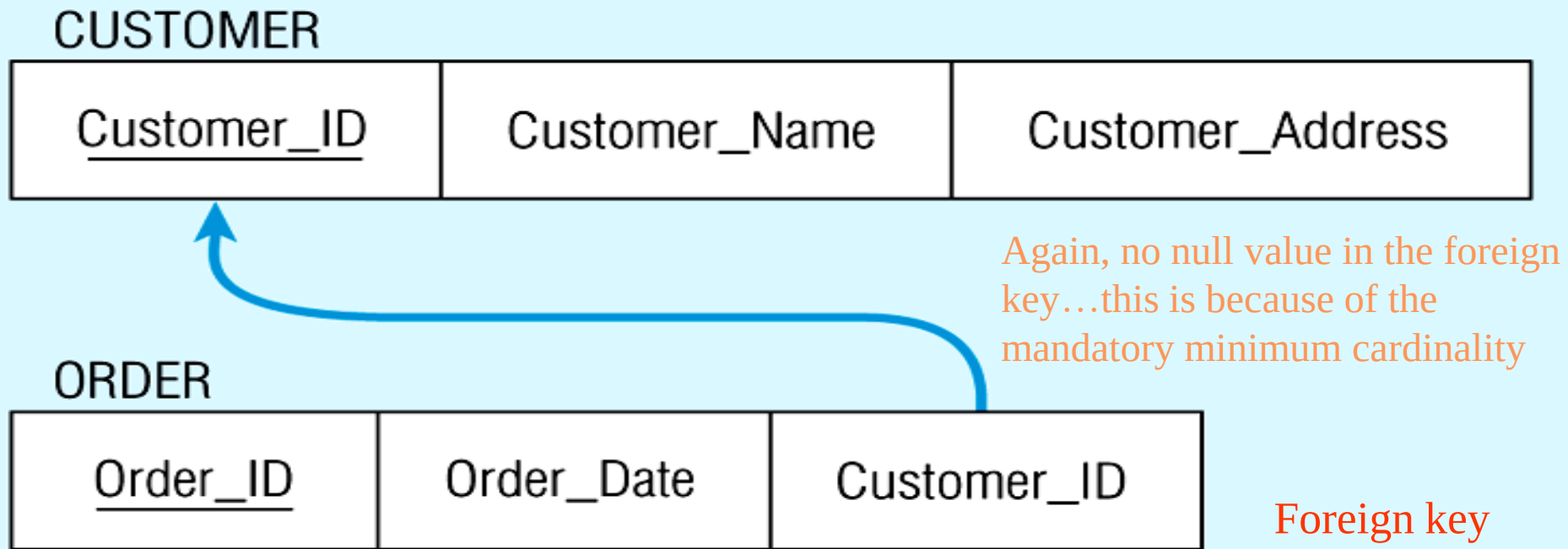


Figure 5-12(b) Mapping the relationship



Associative Entities

Mapping Associative Entities

- Identifier Not Assigned
 - Default primary key for the association relation is composed of the primary keys of the two entities (as in M:N relationship)
- Identifier Assigned
 - It is natural and familiar to end-users
 - Default identifier may not be unique

Figure 5-15: Mapping an associative entity

(a) Associative entity

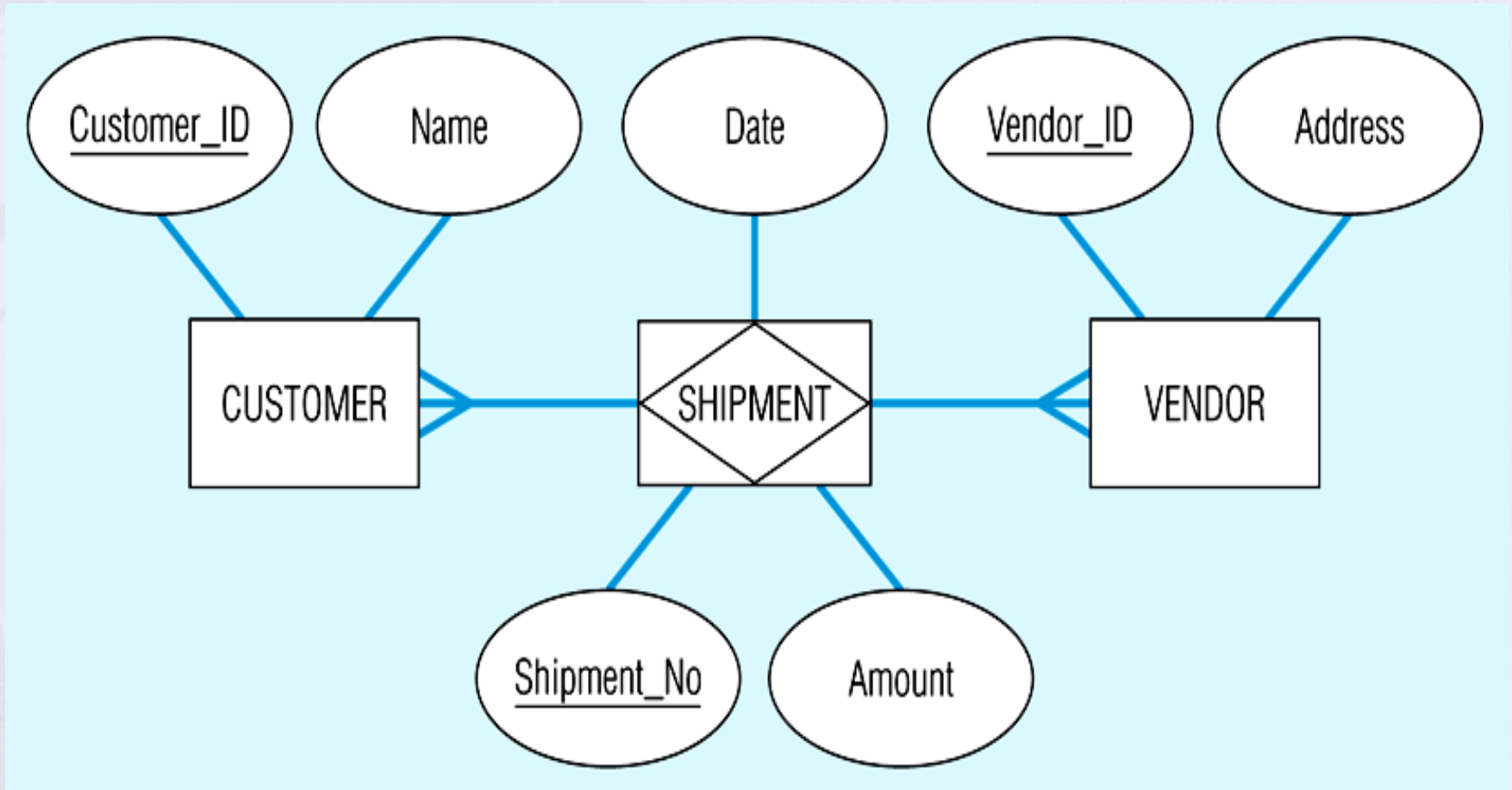


Figure 5-15(b) Three resulting relations

CUSTOMER

<u>Customer_ID</u>	Name	(Other Attributes)
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SHIPMENT

<u>Shipment_No</u>	<u>Customer_ID</u>	<u>Vendor_ID</u>	Date	Amount
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VENDOR

<u>Vendor_ID</u>	Address	(Other Attributes)
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